Google Summer of Code

ML4SCI – HEPSIM GSoC 2026 Evaluation Summary

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Approach

The analysis was performed on five standard `QG_jets*.npz` files, corresponding to 5×10^5 jets, with class balance $N_g = N_q = 2.5 \times 10^5$. For each jet, padded constituents were removed using the constituent mask $m_i = \mathbf{1}[p_{T,i} > 0]$, and the physical multiplicity was computed as $n = \sum_i m_i$. Constituent four-momenta were reconstructed from (p_T, η, ϕ) through

$$p_x = p_T \cos\phi, \qquad p_y = p_T \sin\phi, \qquad p_z = p_T \sinh\eta, \qquad E = p_T \cosh\eta,$$

treating constituents as approximately massless. The jet four-vector was then obtained by $P_J^\mu = \sum_i p_i^\mu = (E_J, \vec{p}_J)$, from which the jet mass was evaluated as $m_J = \sqrt{\max(E_J^2 - \|\vec{p}_J\|^2, 0)}$. Two additional observables were constructed: the jet width

$$w = \frac{\sum_i p_{T,i}\, \Delta R_i}{\sum_i p_{T,i}}, \qquad \Delta R_i = \sqrt{(\eta_i - \eta_J)^2 + (\phi_i - \phi_J)^2},$$

and the transverse-momentum dispersion

$$p_T^D = \frac{\sqrt{\sum_i p_{T,i}^2}}{\sum_i p_{T,i}}.$$

To probe center-of-mass geometry, each jet was boosted to its rest frame using the Lorentz velocity

$$\vec{\beta} = \frac{\vec{p}_J}{E_J}, \qquad \gamma = \frac{1}{\sqrt{1 - \|\vec{\beta}\|^2}},$$

and each constituent four-vector was transformed with the standard Lorentz boost, yielding boosted momenta $p_i'^\mu$ satisfying $\sum_i \vec{p}_i' \approx 0$. Rest-frame engineered features included multiplicity, sphericity-like and aplanarity-like shape measures derived from the momentum tensor, a thrust-like quantity, the maximum constituent energy fraction, and $\langle |\cos\theta| \rangle$. A binary classifier was then trained on these rest-frame features and compared to a lab-frame baseline through ROC/AUC analysis and thresholded classification diagnostics.

Results

The dataset contains 13,288,604 physical constituents in gluon jets and 8,347,536 in quark jets, corresponding to mean multiplicities 53.154 and 33.390, respectively, confirming the expected higher radiation pattern of gluon jets. Mean lab-frame observables also follow established quark–gluon behavior: gluon jets are heavier and broader, while quark jets exhibit larger p_T^D, consistent with momentum being concentrated in fewer, harder constituents. The boost implementation was numerically validated by checking the residual rest-frame momentum, which was found to be at the 10^{-12} to 10^{-13} level for representative jets, confirming $\sum_i \vec{p}_i' \simeq 0$ to machine precision. For classification, the rest-frame model achieved an AUC of 0.85014, while the lab-frame baseline achieved 0.85197, indicating that the chosen rest-frame feature set is physically interpretable and competitive but does not materially outperform the simpler lab-frame baseline. The held-out confusion matrix at the chosen working point,

$$\begin{bmatrix} 37739 & 12212 \\ 10644 & 39405 \end{bmatrix},$$

shows balanced discrimination with a moderate false-positive/false-negative tradeoff. Among all tested single-feature rest-frame variables, constituent multiplicity was the most powerful discriminator with single-feature AUC 0.84098, substantially above other geometric descriptors, implying that global particle-count structure still dominates the separation.

Dataset / Observable Summary	**Classification / Diagnostic Summary**
Total gluon jets: 250,000	Rest-frame AUC: 0.85014
Total quark jets: 250,000	Lab-frame AUC: 0.85197
Total gluon constituents: 13,288,604	Most discriminating feature: multiplicity
Total quark constituents: 8,347,536	Single-feature AUC of multiplicity: 0.84098
Mean multiplicity (g, q): 53.154, 33.390	Rest-frame sphericity AUC: 0.68969
Mean jet mass (g, q): 46.8831, 32.0947	Rest-frame aplanarity AUC: 0.69104
Mean jet width (g, q): 0.0654, 0.0389	Rest-frame thrust-like AUC: 0.70112
Mean p_T^D (g, q): 0.2728, 0.3826	Max energy fraction AUC: 0.77144
Boost check $\|\sum \vec{p}'\|$: $10^{-12} - 10^{-13}$	Mean $\|\cos\theta\|$ AUC: 0.70112

Conclusion. The analysis establishes the full workflow requested in the evaluation: structured loading of padded variable-length jet data, constituent-level counting with padding removal, computation of physically meaningful jet observables, Lorentz boosting into the center-of-mass frame with numerical validation, and supervised quark–gluon discrimination from rest-frame features. The measured trends are physically consistent and the classification study shows that, for this handcrafted feature design, the rest frame provides interpretability and geometric structure but no significant performance gain over a strong lab-frame baseline.